

Title: SERVICE PRISMATIC COMPASS AND TYPES OF BEARINGS

National Cadet Corps (NCC) â Official Directorate General Training Notes

1. THE SERVICE PRISMATIC COMPASS MARK III A: The primary handheld navigational instrument of the Indian Armed Forces. Key components include the Magnetic Needle, Liquid-filled Dial (graduated 0° to 360° clockwise), Glass Window with Hair Line, Magnifying Prism, Thumb Ring, and Clamping Screw.

2. DEFINITION OF A BEARING: A Bearing is the clockwise horizontal angular measurement taken from a fixed zero reference North direction to a target object, expressed in degrees from 000° to 360° with 3-digit notation.

3. THE THREE TYPES OF NORTH: 1. True North (TN): Direction of the geographic North Pole; 2. Grid North (GN): Direction of vertical Easting grid lines on the map; 3. Magnetic North (MN): Direction indicated by the compass needle pointing to the Earth's Magnetic North Pole.

4. MAGNETIC VARIATION (DECLINATION): The angular difference between True North and Magnetic North, varying annually based on planetary geomagnetic drift.